

no ground truth is used; $\log_2$ probes, not a scan

take S of the batch's own queries

answer them at a generous $\alpha_{\max}$
keep that as the reference

bisect the grid: answer the same S at the
midpoint, count top- k slots that differ

run the rest of the batch at α^*

within tolerance:
search below it

not within:
search above it

probe 1

probe 2

probe 3

picked $\alpha^* = 4$

$\alpha = 16$, 0 slots differ

$\alpha = 4$, 0 slots differ

$\alpha = 2$, 3 slots differ

(b) openai3-3072, nprobe = 128, $S = 32$: 3 probes, 56.8 ms

α grid, walked by bisection